

“Fun” Hobby Motor Assignment

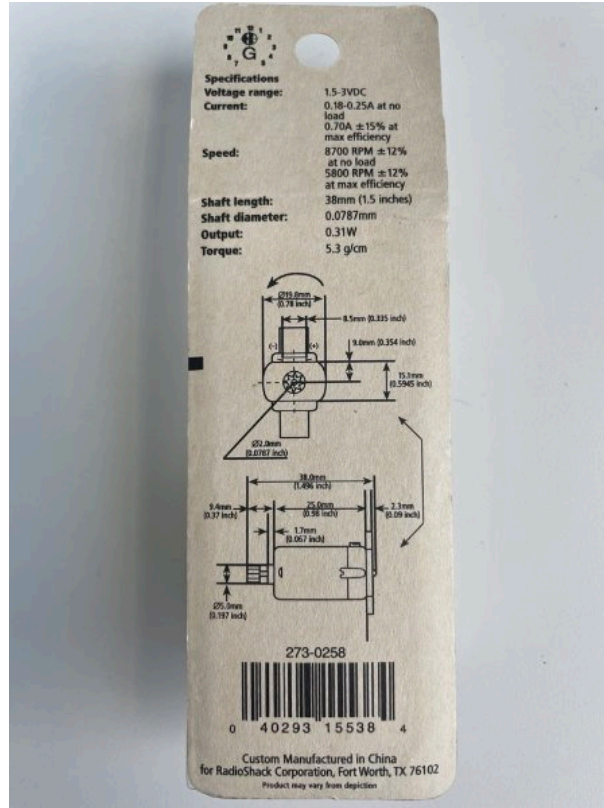


Figure 1: Schematic for RadioShack hobby motor, adapted from “Motor Specs Fun Problem (ESC-251-1-Systems Engineering-2026SP)”.

Note that the following are values given by the problem:

$$v = 3 \text{ V}, \quad I = 0.25 \text{ A}, \quad \omega = 8700 \text{ rpm}, \quad T = (5.3) \frac{\text{g}}{\text{cm}}$$

1.

We can express the differential equation governing this motor with:

$$v = L\dot{I} + RI + e_b \quad (1)$$

where v is the input voltage to the motor, and e_b is the back-emf of the motor.

When no load is applied to the motor, $((\dot{I} = 0) \wedge (I \approx 0))$, therefore Equation 1 can be rewritten as:

$$v = e_b$$

We can also express the back-emf of the system with:

$$e_b = K_b \omega \quad (2)$$

Substituting Equation 2 into Equation 1 yields:

$$v = K_b \omega \quad (3)$$

Rearranging Equation 3 yields an equation for K_b , which can be simplified with the corresponding givens:

$$K_b = \frac{v}{\omega} \approx (3.448 * 10^{-4}) \frac{V}{\text{rpm}} \quad (4)$$

2.

If we were to take the Laplace Transform of both sides of Equation 3, we get:

$$V(s) = K_b \Omega(s) \quad (5)$$

We can rearrange Equation 5 to get the system's transfer function:

$$\frac{\Omega(s)}{V(s)} = \frac{1}{K_b} \quad (6)$$

The system's DC gain can be expressed as:

$$\left(\lim_{s \rightarrow 0} \right) \left(\frac{\Omega(s)}{V(s)} \right) = \left(\lim_{s \rightarrow 0} \right) \left(\frac{1}{K_b} \right) = \frac{1}{K_b} = (2900) \frac{\text{rpm}}{V} \quad (7)$$

3.

Now, let the input voltage ($v_i = (U(t))$ V), where $U(t)$ is the Heaviside function. Assume that ($t \in \mathbb{R}_{\geq 0}$). Therefore, v_i can be rewritten as ($v_i = 1V$).

We can express the steady-state motor speed ω_{ss} as:

$$w_{ss} = \left(\left(\lim_{s \rightarrow 0} \right) \left(\frac{\Omega(s)}{V(s)} \right) \right) (v_i) = 2900 \text{ rpm} \left(\frac{V}{V} \right) = 2900 \text{ rpm} \quad (8)$$